

K SHREEKRISHNA UPADHYAYA

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PROFILE

Curious engineering student focused on low-level development, systems programming, software architecture, and interested in AI. Learning through experimentation and hands-on projects while exploring modern software development workflows.

EDUCATION

VIVEKANANDA COLLEGE OF ENGINEERING AND TECHNOLOGY, PUTTUR

DK, Karnataka

Bachelor of Engineering in Artificial Intelligence & Machine Learning

September 2023 – Present

Cumulative GPA (till 6th semester): 8.69/10.0

Relevant Coursework: Data Structures, OS, Computer Networks, AI, Machine Learning

ACTIVITIES & CONTRIBUTIONS

Hash Shell | C

January 2026 - Present

Contributor

- Optimized interactive shell behavior by resolving multi-line continuation bugs in quoted strings.
- Fixed tab-completion prefix errors and ensured consistent terminal state during interactive mode.
- Refactored long functions and aligned code with the coding standards according to contributing guidelines.
- Adhered to upstream coding standards and performance requirements during the PR review process.

PROJECTS

Sn Project | C, x86_64 Assembly

December 2025 - Present

- Developing a modular collection of cross-platform C libraries that provide reusable building blocks for native software development.
- Implemented reusable components including virtual memory abstraction, file system APIs, threading, synchronization, logging, tracing, and dynamic loading with consistent APIs and minimal dependencies.

FinDiary | Go, PostgreSQL, Flutter, gRPC

July 2026 - Present

- Developing a personal finance tracking application using a Specification-Driven Development (SDD) workflow.
- Building backend services in Go with PostgreSQL and a Flutter-based client application.
- Using agentic AI to generate implementation from reviewed specifications while maintaining architectural consistency.

Snuk | C

April 2023 - Present

- Developing an embeddable scripting language for integration into native applications.
- Exploring language design, lexical analysis, parsing, interpreter implementation, and lightweight runtime systems for embedding into native applications.

Regex Engine | C

December 2025

- Built a regular expression parser supporting grouping, alternation, concatenation, and quantifiers.
- Implemented Thompson-style NFA construction and simulation for efficient pattern matching.

LC3VM & Kilo Editor | C

2024 – 2025

- Implemented an LC-3 virtual machine including instruction decoding, memory model, and execution loop.
- Built a terminal-based text editor implementing cursor movement, text buffers, and editing operations.

Other Projects:

- **Systems & Tools:** StateFlow (DFA/NFA simulator), whcli (Wallhaven API client), WSO (web shortcut manager, shell).
- **Backend & Networking:** GoAuth (Go REST service), WOL-Android (Wake-on-LAN utility).

SKILLS & OTHERS

Languages & Build Tools: C, C++, CMake, Python, x86_64 Assembly(inline/MASM), Go, Java, POSIX Shell Script;

Systems: POSIX APIs, Memory Management, Virtual Memory, Threading, Synchronization, File Systems;

Backend & Development: REST APIs, PostgreSQL, Flutter, Specification-Driven Development, Agentic AI Workflows;

Tools: Git, Linux/Unix Shell, Typst, Vim;

Awards: Winner of **Nexathon 2025** (SDIT) and **DevHack 2025** (Sahyadri College).